

#DAY 6: SQL & Microservices:

1. AND

Definition: Returns rows only when all conditions are true.

Example:

```
SELECT * FROM Employees  
WHERE Department = 'IT' AND Salary > 50000;
```

2. OR

Definition: Returns rows when at least one condition is true.

Example:

```
SELECT * FROM Employees  
WHERE Department = 'IT' OR Department = 'HR';
```

3. NOT

Definition: Reverses the condition.

Example:

```
SELECT * FROM Employees  
WHERE NOT Department = 'Sales';
```

4. IN

Definition: Checks whether a value exists in a list.

Example:

```
SELECT * FROM Employees  
WHERE Department IN ('IT', 'HR');
```

5. BETWEEN

Definition: Checks whether a value is within a range (inclusive).

Example:

```
SELECT * FROM Employees  
WHERE Salary BETWEEN 30000 AND 50000;
```

6. LIKE (Pattern Matching)

Definition: Searches for values that match a specific pattern.

Example:

```
SELECT * FROM Employees  
WHERE Name LIKE 'A%';
```

(Returns names starting with A.)

7. IS NULL

Definition: Checks whether a column contains a NULL value.

Example:

```
SELECT * FROM Employees  
WHERE Manager_ID IS NULL.
```

8. WHERE

Definition: Filters rows before grouping.

Example:

```
SELECT * FROM Employees  
WHERE Salary > 50000.
```

9. GROUP BY

Definition: Groups rows with the same values and is used with aggregate functions.

Example:

```
SELECT Department, COUNT(*)  
FROM Employees  
GROUP BY Department;
```

10. HAVING

Definition: Filters grouped data after GROUP BY.

Example:

```
SELECT Department, COUNT(*)  
FROM Employees  
GROUP BY Department  
HAVING COUNT(*) > 5;
```

11. UNION

Definition: Combines results of two queries and removes duplicate rows.

Example:

```
SELECT Name FROM Students  
UNION  
SELECT Name FROM Teachers;
```

12. UNION ALL

Definition: Combines results of two queries and keeps duplicate rows.

Example:

```
SELECT Name FROM Students  
UNION ALL  
SELECT Name FROM Teachers;
```

13. INTERSECT

Definition: Returns only the common rows from two queries.

Example:

```
SELECT Name FROM Students  
INTERSECT  
SELECT Name FROM Teachers;
```

14. INNER JOIN

Definition: Returns only matching rows from both tables.

Example:

```
SELECT e.Name, d.DepartmentName  
FROM Employees e  
INNER JOIN Departments d  
ON e.DeptID = d.DeptID;
```

15. LEFT JOIN

Definition: Returns all rows from the left table and matching rows from the right table.

Example:

```
SELECT e.Name, d.DepartmentName  
FROM Employees e  
LEFT JOIN Departments d  
ON e.DeptID = d.DeptID;
```

16. RIGHT JOIN

Definition: Returns all rows from the right table and matching rows from the left table.

Example:

```
SELECT e.Name, d.DepartmentName  
FROM Employees e  
RIGHT JOIN Departments d  
ON e.DeptID = d.DeptID.
```

17. Microservices

Definition: A software architecture where an application is divided into small, independent services, each performing a specific business function and communicating through APIs.

Example:

An e-commerce application can have:

User Service

Product Service

Order Service

Payment Service

Each service works independently and communicates with the others through APIs.